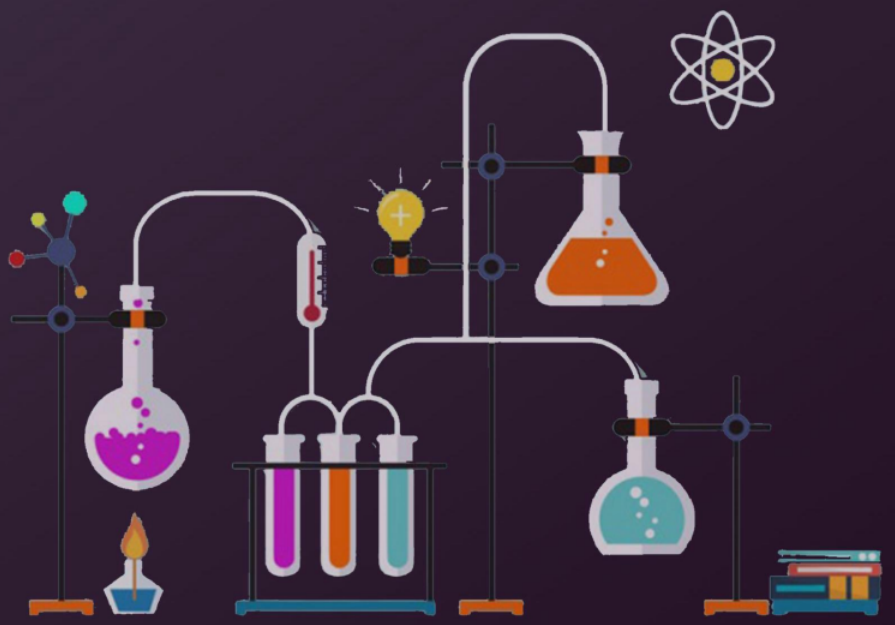
CHEMISTRY

Oxidation Reduction



SHEET

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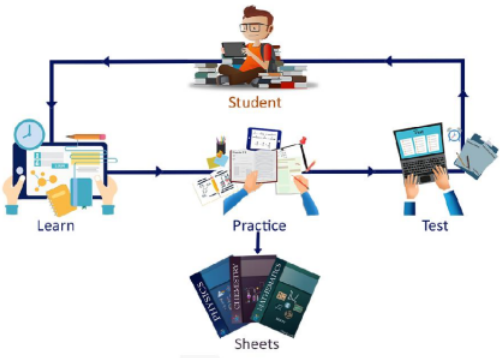
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr. No.	Topics	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Recap of Oxidation and Oxidation of Diols	1	7	1	
2	Oxidation of Alcohols	2-3	1, 8		
3	Distinguishing Tests Between Aldehyde and Ketone	3-8	2-3	2-5	1-2, 5
4	Reduction by Hydrogenation, Rosenmund's and Stephen's Reduction	9-12		6	
5	LAH, SBH and Bouevault Blanc Reduction	13-15	9	7-11	6-8
6	Wolff Kishner and Clemmensen Reduction			12	9
7	Mozingo Reduction, Red P and HI Reduction	16			
8	ACC	17	4-6, 10	13-15	3-4

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. On catalytic reduction with H_2/Pt how many alkenes will give n-butane ? (Including stereoisomers)

(A) 1 (B) 2

(C) 3 (D) 4

Oxidation of Alcohols

Q2. Which one of the following alcohols are oxidised by MnO_2?

(A) $C_6H_5 - CH_2 - CH_2 - OH$

(B) $CH_2 = CH - CH_2 - \underset{\displaystyle OH}{\underset{|}{C}}H - CH_3$

(C) $CH_3 - CH = CH - \underset{\displaystyle OH}{\underset{|}{C}}H - CH_3$

(D) $CH_3 - CH_2 - CH_2 - OH$

Q3. Consider the following sequence of reactions:-

$$CH_3CHO \xrightarrow[2.\ H_3O^+]{1.\ CH_3MgBr} A \xrightarrow[H_2SO_4]{K_2Cr_2O_7} B$$

The product (B) is-

(A) CH_3COOCH_3 (B) $CH_3CHOHCH_3$

(C) CH_3COCH_3 (D) CH_3COOH

Distinguishing Tests Between Aldehyde and Ketone

Q4. Silver mirror with Tollen's reagent is given by the compound :-

(A) Benzaldehyde (B) Methanal

(C) Methanoic acid (D) All the above

Q5. Glucose gives test with :-

(A) Tollen's reagent

(B) Fehling solution

(C) Benedict's solution

(D) All

Q6. To distinguish between formic acid and acetic acid, suitable reagent is :

(A) Tollen's reagent

(B) Ethanol

(C) Litmus solution

(D) Sodium bicarbonate

Q7. $CH_3 - \overset{\displaystyle O}{\overset{\|}{C}} - H$ and $CH_3 - \overset{\displaystyle O}{\overset{\|}{C}} - CH_3$ is differentiated by -

(A) Tollen's reagent

(B) Lucas test

(C) Iodoform test

(D) $NaHSO_3$ test

Q8. Which will give silver mirror test with Tollens reagent :

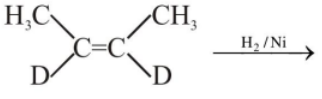
(A) C_6H_5CHO

(B) CH_3–CHO

(C) HCOOH

(D) All of these

Reduction by Hydrogenation, Rosenmund's and Stephen's Reduction

Q9. 

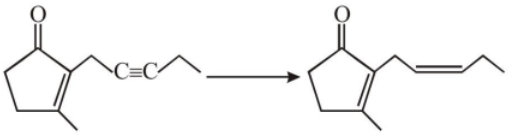
Product of above reaction will be :

(A) Racemic mixture

(B) Diastereomers

(C) Meso

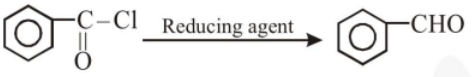
(D) Constitutional isomers

Q10. 

Which reagent will be used for the above conversion ?

(A) Na/Liq. NH_3 (B) H_2, Pd–$CaCO_3$

(C) Li, Ph–NH_2 (D) H_2, Pt

Q11. 

Reducing agent is -

(A) Red P and HI

(B) Pd + $BaSO_4$ / H_2

(C) $LiAlH_4$ / H^+

(D) Zn–Hg / conc. HCl

Q12. $CH_3-C\equiv N \xrightarrow[+HCl]{SnCl_2} A \xrightarrow[H_2O]{H^\oplus} B \xrightarrow[Ni]{H_2} C$

'C' is -

(A) CH_3-CH_2-OH

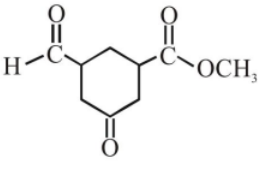
(B) CH_3-COOH

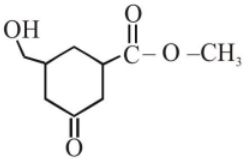
(C) CH_3-CHO

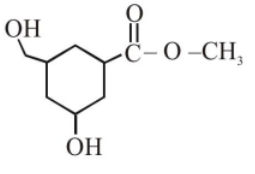
(D) $CH_3-CH_2-NH_2$

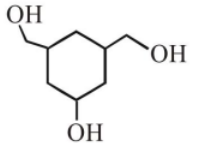
LAH, SBH and Bouevault Blanc Reduction

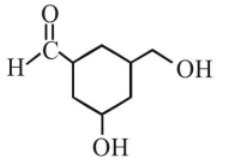
Q13. Predict the product when given compound reacts with $LiAlH_4$:-

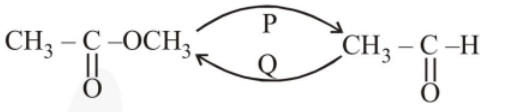


(A) 

(B) 

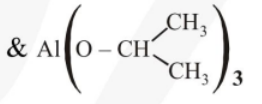
(C) 

(D) 

Q14. 

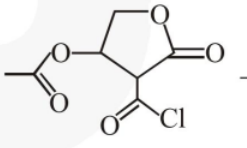
P & Q respectively is -

(A) DIBAL–H followed by H_2O & $Al(OCH_2CH_3)_3$

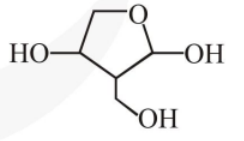
(B) $LiAlH_4$ followed by H_2O & 

(C) DIBAL–H followed by H_2O & m–CPBA followed by CH_3OH/H^+

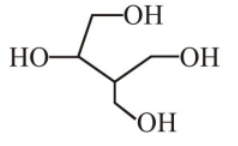
(D) None of these

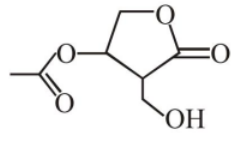
Q15.  $\xrightarrow{NaBH_4}$ Product

The product is :

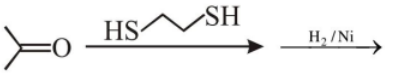
(A) 

(B) CH_3-CH_2-OH

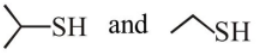
(C) 

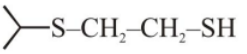
(D) 

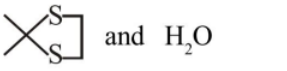
Mozingo Reduction, Red P and HI Reduction

Q16. 

The end products of the reaction are :

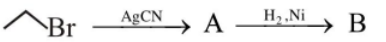
(A) 

(B) 

(C) 

(D) 

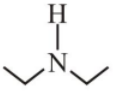
ACC

Q17. 

The final product (B) is :

(A) 

(B) 

(C) 

(D) 

Exercise - 1A — *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Which one of the following alcohols can be oxidised by K_2CrO_4?

(A) Ethanol

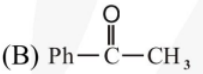
(B) Tert butyl alcohol

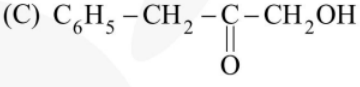
(C) Isopropyl alcohol

(D) Allyl alcohol

Q2. Silver mirror test with Tollens reagent is given by :

(A) C_6H_5CHO

(B) 

(C) 

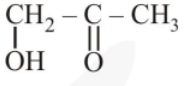
(D) CH_3CHO

Q3. Fehling solution gives red precipitate with:

(A) Aromatic aldehyde

(B) Aliphatic aldehyde

(C) Ketones

(D) 

MATCHING LIST TYPE

Each question has TWO (02) ***matching lists: LIST I and LIST II****. FOUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

Q4. Match the given reaction in List I with the reagent used in List II

List – I (Reaction)	**List – II (Reagent)**
(P) 1-(3-Hydroxycyclopentyl) ethanone → 3-Ethylcyclopentanol	(1) NH_2-NH_2 /OH^-/glycol
(Q) Cyclohexanone → cyclohexanol	(2) DIBAL-H
(R) Cyclohexane carbonyl chloride → Cyclohexane carbaldehyde	(3) $LiAlH_4$
(S) Phenylbenzoate → Benzaldehyde	(4) H_2/Pd/$BaSO_4$/Quinoline

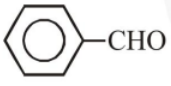
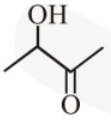
Code :

	P	Q	R	S
(A)	1	2	4	3
(B)	1	3,2	4,2	2
(C)	1	2	4,2	3
(D)	4	2	1	3

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

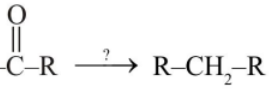
Q5.

Column - I	**Column - II**
(A) 	(P) Positive Iodoform test
(B) CH_3–CHO	(Q) Reduces Fehling solution
(C) HCHO	(R) Positive Tollen's test
(D) 	(S) Brady's reagent turns red

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

REMOVE

Q6. 

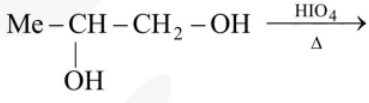
Identify numbers of reagent that can be used for above conversion.

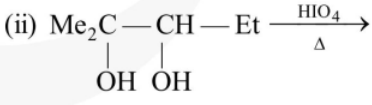
(A) Zn–Hg / HCl (B) $LiAlH_4$

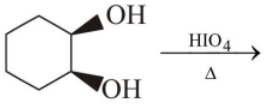
(C) $CHCl_3$ + NaOH (D) N_2H_4 / $O^{\ominus}H$

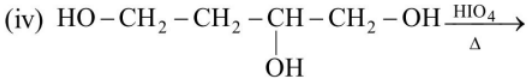
(E)  , H_2 / Ni (F) SeO_2

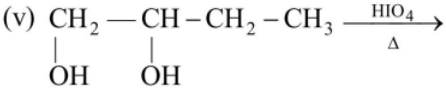
SUBJECTIVE TYPE

Q7. (i) 

(ii) 

(iii) 

(iv) 

(v) 

(vi) $CH_2(OH)-CH(OH)-CH(OH)-CH_3 \xrightarrow[\Delta]{HIO_4}$

(vii) $CH_2(OH)-CH(OH)-CH(OH)-CH_2(OH) \xrightarrow[\Delta]{HIO_4}$

(viii) $Me-C(=O)-CH(OH)-Me \xrightarrow[\Delta]{HIO_4}$

(ix) $Me-C(=O)-C(=O)-Me \xrightarrow[\Delta]{HIO_4}$

Q8. (i) $CH_3-CH_2-CH_2-OH \xrightarrow[\text{or } KMnO_4/H^{\oplus},\Delta]{KMnO_4/OH^{-},\Delta} ?$

(ii) $CH_3-CH(OH)-CH_2-CH_3 \xrightarrow[\text{or } K_2Cr_2O_7,H^{\oplus}]{KMnO_4,H^{\oplus}} ?$

$\xrightarrow[\text{or conc. } HNO_3,\Delta]{K_2Cr_2O_7/H^{\oplus},\Delta} ?$

Q9. Identify the product ?

(i) $Me-CHO \xleftarrow{NaBH_4}$ / $\xrightarrow{LiAlH_4}$ / $\xrightarrow{DIBAL\text{-}H}$

(ii) $Me_2CO \xleftarrow{NaBH_4}$ / $\xrightarrow{LiAlH_4}$ / $\xrightarrow{DIBAL\text{-}H}$

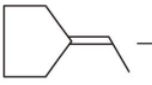
(iii) $Me-COCl \xleftarrow{NaBH_4}$ / $\xrightarrow{LiAlH_4}$ / $\xrightarrow{DIBAL\text{-}H}$

(iv) $Me-COOEt \xleftarrow{NaBH_4}$ / $\xrightarrow{LiAlH_4}$ / $\xrightarrow{DIBAL\text{-}H}$

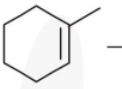
Q10. (i) $CH_2=CH_2 \xrightarrow{H^{\oplus}/KMnO_4}$

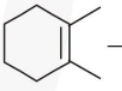
(ii) $CH_3-CH=CH_2 \xrightarrow{H^{\oplus}/KMnO_4}$

(iii)  $\xrightarrow{H^{\oplus}/KMnO_4}$

(iv)  $\xrightarrow{H^{\oplus}/KMnO_4}$

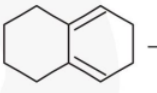
(v)  $\xrightarrow{H^{\oplus}/KMnO_4}$

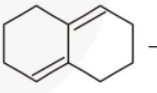
(vi)  $\xrightarrow{H^{\oplus}/KMnO_4}$

(vii)  $\xrightarrow{H^{\oplus}/KMnO_4}$

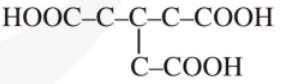
(viii)  $\xrightarrow[-CO_2]{H^{\oplus}/KMnO_4}$

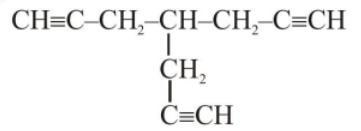
(ix)  $\xrightarrow{H^{\oplus}/KMnO_4}$

(x)  $\xrightarrow{H^{\oplus}/KMnO_4}$

(xi)  $\xrightarrow{H^{\oplus}/KMnO_4}$

(xii) $C_{10}H_{10} \xrightarrow{H^{\oplus}/KMnO_4}$

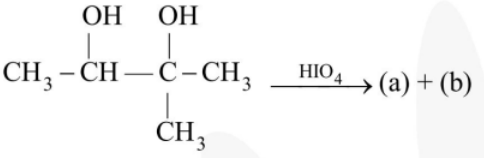


(xiii) 

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. In the given reaction :



(a) and (b) respectively be :

(A) CH_3CHO and CH_3CHO

(B) CH_3COCH_3 and CH_3CHO

(C) CH_3COCH_3 and CH_3COCH_3

(D) CH_3COOH and CH_3COCH_3

Q2. The reagent that can distinguish benzaldehyde from propionaldehyde is -

(A) I_2/NaOH

(B) Fehling's solution

(C) Tollens reagent

(D) 2, 4-dinitrophenyl hydrazine

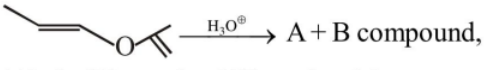
Q3. Which of the following do not reduce Fehling solution.

(1) CH_3CHO (2) Ph–CHO

(3) HCOOH (4) HCHO

(A) 1 Only (B) 2,4 Only

(C) 2,3,4 (D) None

Q4.  A + B compound,

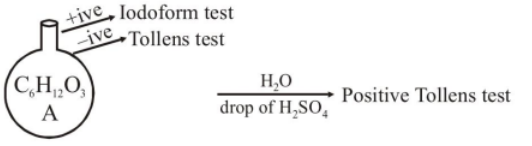
(A) & (B) can be differentiated by -

(A) 2,4–DNP

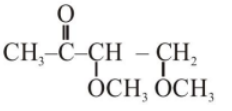
(B) Fehling solution

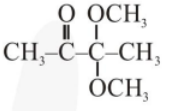
(C) Lucas reagent

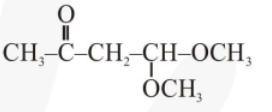
(D) $NaHSO_3$

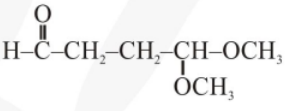
Q5. 

Compound (A) is :-

(A) 

(B) 

(C) 

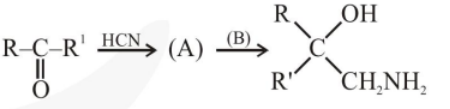
(D) 

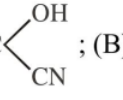
Q6. On catalytic reduction with H_2/Pt how many alkenes will give Isobutane ?

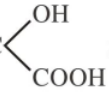
(A) 1 (B) 2

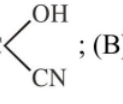
(C) 3 (D) 4

Q7. (A) and (B) in the following reactions are :



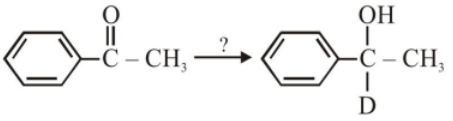
(A) (A) =  ; (B) = $LiAlH_4$

(B) (A) =  ; (B) = NH_3

(C) (A) =  ; (B) = H_3O^+

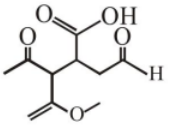
(D) (A) = $RR'CH_2CN$; (B) = NaOH

Q8. Which of the following reagents would carry out the following transformation ?



(A) $NaBD_4$ in CH_3OH
(B) $LiAlH_4$, then D_2O
(C) $NaBD_4$ in CH_3OD
(D) $LiAlH_4$, then D_2O

Q9.



In the above given compound how many functional group reduced by LAH (Lithium aluminium hydride) and SBH (sodium borohydride) respectively?

(A) 4, 4 (B) 4, 3
(C) 3, 4 (D) 4, 2

Q10. An optically active compound 'X' has molecular formula $C_4H_8O_3$. It evolves CO_2 with $NaHCO_3$. 'X' reacts with $LiAlH_4$ to give an achiral compound 'X' is :

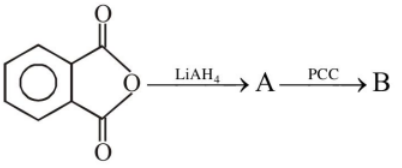
(A) $CH_3CH_2\underset{\underset{OH}{|}}{C}HCOOH$

(B) $CH_3\underset{\underset{Me}{|}}{C}HCOOH$

(C) $CH_3\underset{\underset{CH_2OH}{|}}{C}HCOOH$

(D) $CH_3CH\underset{\underset{OH}{|}}{C}H_2COOH$

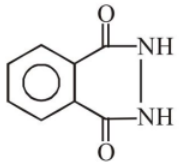
Q11.



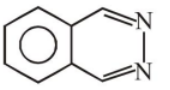
$\xrightarrow{NH_2-NH_2}$ C,

Product (C) is :

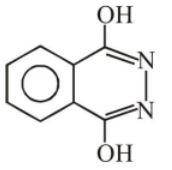
(A)



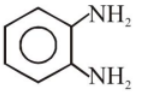
(B)



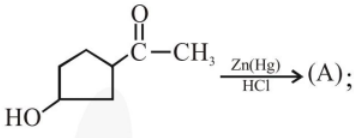
(C)



(D)

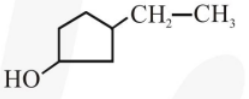


Q12.

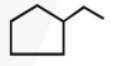


Identify A :-

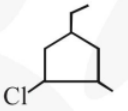
(A)



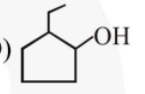
(B)



(C)

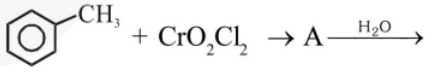


(D)

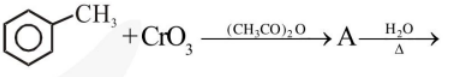


Q13. Benzaldehyde can be formed by :-

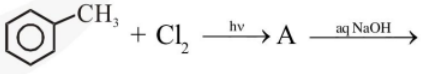
(A)



(B)



(C)



(D) All of them

Q14. 3-Pentanol is produced by which of the following reactions ?

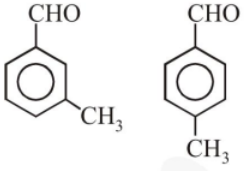
(i) $CH_3CH_2Br \xrightarrow[Ether]{Mg} \xrightarrow{CH_3CH_2CHO} \xrightarrow[H_2O]{H^\oplus}$

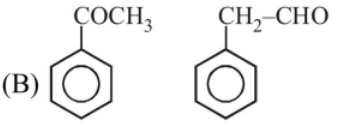
(ii) $HC\equiv CH \xrightarrow{NaNH_2} \xrightarrow{CH_3CH_2CHO} \xrightarrow[H_2O]{H^\oplus}$

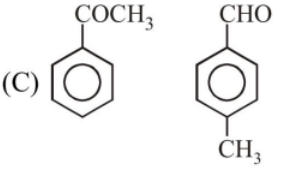
(iii) $(CH_3CH_2)_2CO \xrightarrow[Ether]{LiAlH_4} \xrightarrow[H_2O]{H^\oplus}$

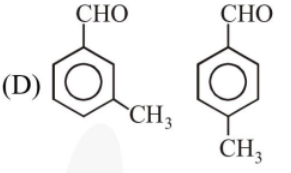
(A) (i) only
(B) (i) & (ii) only
(C) (i) & (iii) only
(D) (ii) & (iii) only

Q15. Out of five carbonyl aromatic isomers of molecular formula containing C_8H_8O. One of the isomer X can give iodoform test while Y can give Benedict's test. What are the structure of X & Y respectively.

(A) 

(B) 

(C) 

(D) 

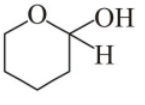
Exercise - 2A *Brain Booster*

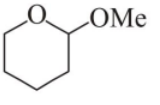
ONE OR MORE THAN ONE CORRECT TYPE

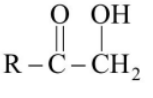
*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

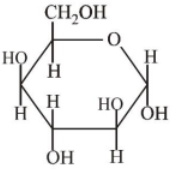
Q1. Schiff's reagent gives pink colour with :
(A) Acetaldehyde
(B) Formic acid
(C) Acetic acid
(D) Methyl acetate

Q2. Which will give the Tollen test.

(A) 

(B) 

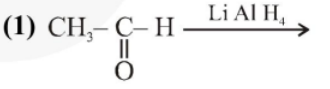
(C) 

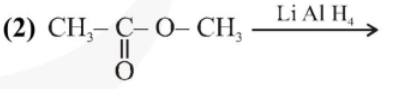
(D) 

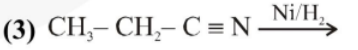
NUMERICAL TYPE

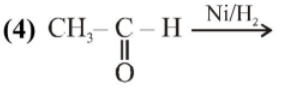
*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

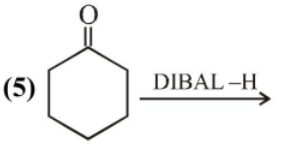
Q3. **Which of the following reaction will form alcohol ?**

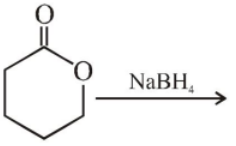
(1) 

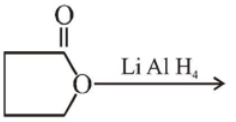
(2) 

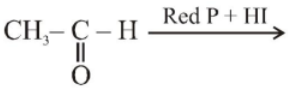
(3) 

(4) 

(5) 

(6) 

(7) 

(8) 

Q4. Number of reagents which will reduce Ph–CH=O into alcohol as one of the final major product(s) followed by acidification -

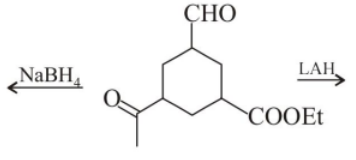
(a) conc. NaOH (b) H_2/Pd
(c) $LiAlH_4$ (d) $NaBH_4$
(e) Zn-Hg/HCl (f) 9-BBN
(g) BH_3/THF (h) Red P/HI
(i) Al $(-OCHMe_2)_3$ / $HOCHMe_2$
(j) CH_3MgX

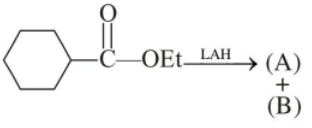
SUBJECTIVE TYPE

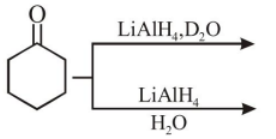
Q5. Identify the product ?

(1) $\xleftarrow{NaBH_4}$ Me–COOH $\xrightarrow{LiAlH_4}$
(2) $\xleftarrow{NaBH_4}$ Me–COOMe $\xrightarrow{LiAlH_4}$
(3) $\xleftarrow{NaBH_4}$ Me–$CONH_2$ $\xrightarrow{LiAlH_4}$
(4) $\xleftarrow{NaBH_4}$ Me–CONH–Me $\xrightarrow{LiAlH_4}$
(5) $\xleftarrow{NaBH_4}$ Me–$CONMe_2$ $\xrightarrow{LiAlH_4}$

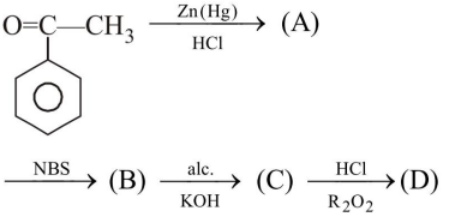
Q6. Give product in following reactions.

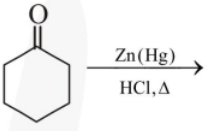
(i) 

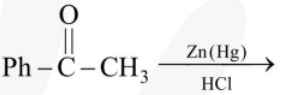
(ii) 

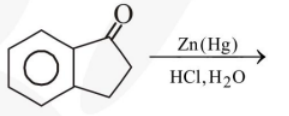
(iii) 

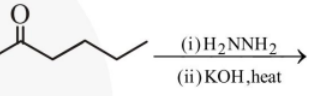
Q7. Solve the following Reaction Sequences:-

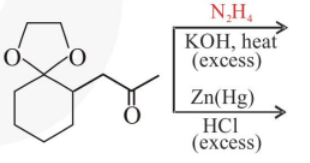
(a) 

(b) 

(c) 

(d) 

(e) 

(f) 

Answer Key

Ex-1									
1	2	3	4	5	6	7	8	9	10
C	C	C	D	D	A	A	D	C	B
11	12	13	14	15	16	17			
B	A	C	C	D	D	B			

Ex - 1A

1. A, C, D **2.** A, C, D **3.** B, D **4.** B

5. (A)→R, S ; (B)→P, Q, R, S ; (C)→Q, R, S ; (D)→P, Q, R, S

6. 3

Ex-2									
1	2	3	4	5	6	7	8	9	10
B	B	B	B	C	A	A	A	B	C
11	12	13	14	15					
B	B	D	C	B					

Ex - 2A

1. A, B **2.** A, C, D **3.** 1, 2, 4, 5, 7 **4.** 8